

First-order approximation algorithms

Weekend 1, Friday morning

Carlos Cotrini

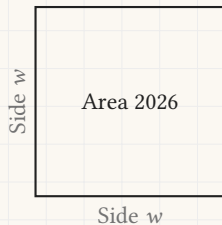
Friday 4 September 2026

1

Section 1

Two problems, done by hand

Calculating square roots

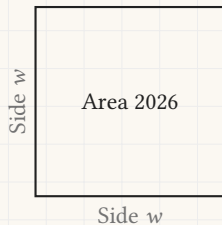


How long is the side?

Find w with $w^2 = 2026$.

w	w^2	$l = 2026 - w^2$	Update
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Calculating square roots

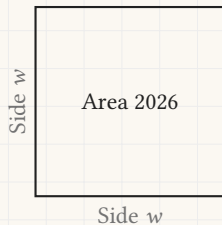


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40	1600	+426	↑

Calculating square roots

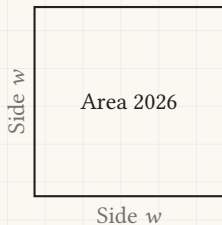


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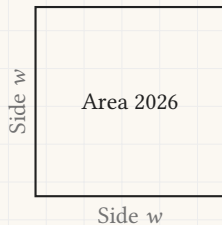


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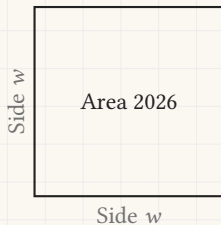


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Nobody here computed a square root. Each row is a guess that was scored, and the score said which way to move.

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$l = 4 - f(w)$	Update
-----	--------	----------------	--------

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$l = 4 - f(w)$	Update
0	-2	+6	↑

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6	124	-120	↓
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⋮	⋮	⋮	

The same four columns, and only f changed.

Written another way,

$$w^3 - 3w^2 + 3w - 2 = (w - 1)^3 - 1,$$

so the equation asks for $(w - 1)^3 = 5$: a cube root wearing a cubic.

The algorithm

```
def solve(f, target, w, alpha):  
    l = target - f(w)           # the verdict: how far off, and which way  
    while abs(l) > 0.001:      # not close enough yet  
        w = w + alpha * l      # the update: move that way  
        l = target - f(w)      # score the new guess  
    return w
```

The algorithm

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def solve(f, target, w, alpha):  
    l = target - f(w)           # the verdict: how far off, and which way  
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```

The two problems of the last ten minutes are the same call, with a different f:

```
def square(w):  
    return w**2  
  
def cubic(w):  
    return w**3 - 3*w**2 + 3*w - 2  
  
solve(square, 2026, 40, 0.005)  
solve(cubic, 4, 0, 0.05)
```


The algorithm

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def solve(f, target, w, alpha):  
    l = target - f(w)           # the verdict: how far off, and which way  
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```

Everything on this slide is settled except α . That is the next section.

2

Section 2

Iterative problem solving

Iterative problem solving

Iterative problem solving

Guess

Guess A value of w , however crude.

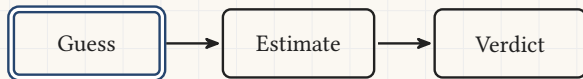
Iterative problem solving



Guess A value of w , however crude.

Estimate What the guess predicts.

Iterative problem solving

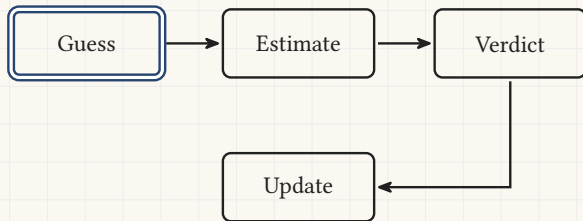


Guess A value of w , however crude.

Estimate What the guess predicts.

Verdict The signed miss l .

Iterative problem solving



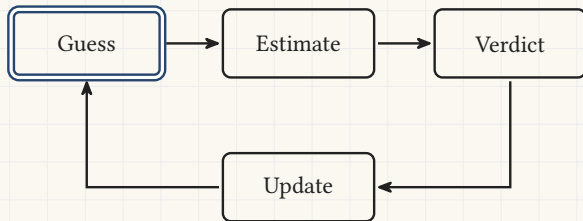
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Verdict The signed miss l .

Update Move w the way the verdict says.

Iterative problem solving



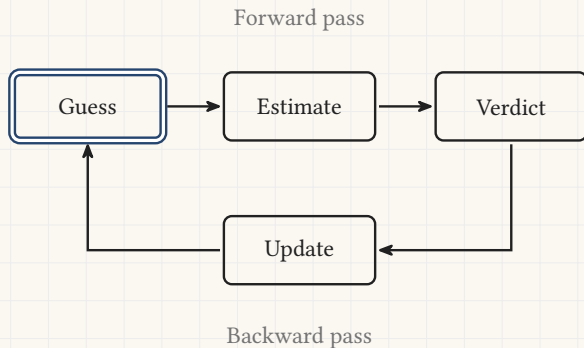
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Iterative problem solving



Guess A value of w , however crude.

Estimate What the guess predicts.

Verdict The signed miss l .

Update Move w the way the verdict says.

The update rule

$$w := w$$

w	w^2	l
-----	-------	-----

w	w^2	l
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The update rule

$$w := w + \underbrace{l}_{\text{The verdict}}$$

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$$w := w + \underbrace{\alpha}_{\text{How far to move}} \underbrace{l}_{\text{The verdict}}$$

w	w^2	l
40.000	1600.000	426.000
42.130	1774.937	251.063
43.385	1882.286	143.714
44.104	1945.153	80.847
44.508	1980.973	45.027
44.733	2001.064	24.936
44.858	2012.234	13.766
44.927	2018.414	7.586
44.965	2021.824	4.176
44.986	2023.702	2.298
44.997	2024.736	1.264
45.003	2025.305	0.695

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45.011	2025.981	0.019
45.011	2025.989	0.011
45.011	2025.994	0.006
45.011	2025.997	0.003
45.011	2025.998	0.002
45.011	2025.999	0.001

- Starting at $w = 40$, with $\alpha = 0.005$.
- The whole run, all 23 rounds of it.
- From the fifth row of this block, w has stopped moving and the verdict is still shrinking.

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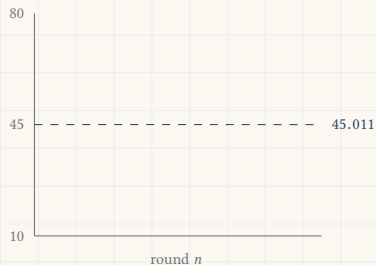
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- Starting at $w = 40$, with $\alpha = 0.005$.
- The whole run, all 23 rounds of it.
- From the fifth row of this block, w has stopped moving and the verdict is still shrinking.
- $\sqrt{2026} = 45.011110$, and no square root was ever taken.

α is the step size. The letter ε is reserved: it is the stopping tolerance, and at 11:00 it becomes the width of a difference window.

Letting the step shrink

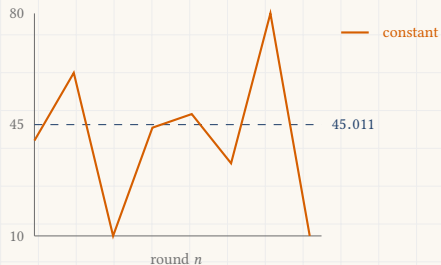
$w := w + \alpha_n l$, with $\alpha_n = 0.05/n$: the step gets smaller every round.



n	α_n	w	l
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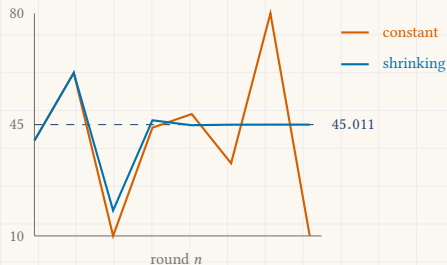


The constant run leaves the picture twice.

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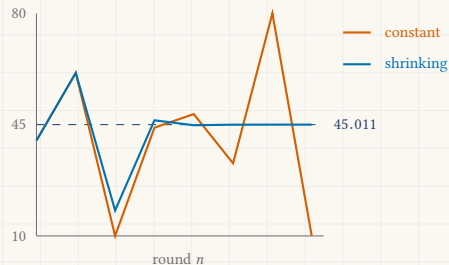


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n	α_n	w	l
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2	0.0250	61.300	-1731.690
3	0.0167	18.008	+1701.721
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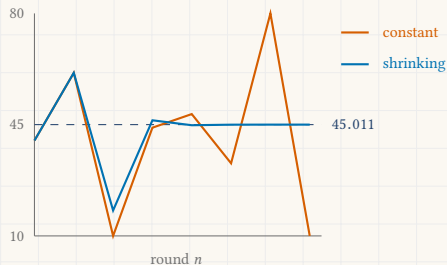


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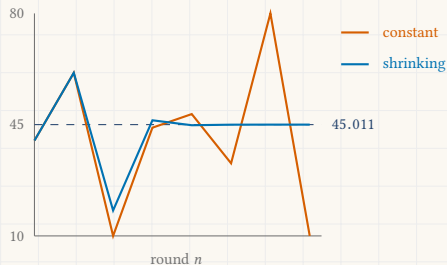


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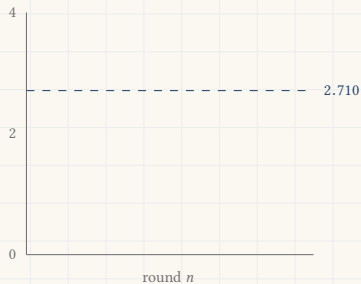
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$\sqrt{2026} = 45.011110$. The digits arrive one at a time, and no square root was ever taken.

Letting the step shrink: the equation

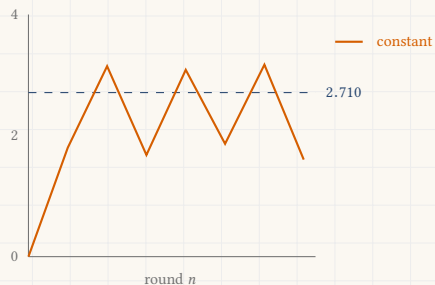
The same change, on $w^3 - 3w^2 + 3w - 2 = 4$, with $\alpha_n = 0.3/n$.



n	α_n	w	l
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Letting the step shrink: the equation

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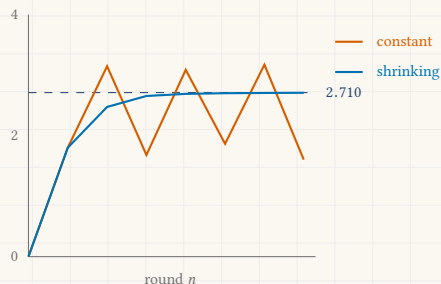


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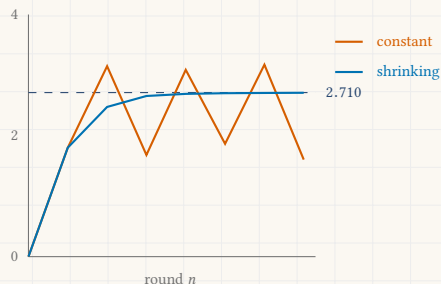


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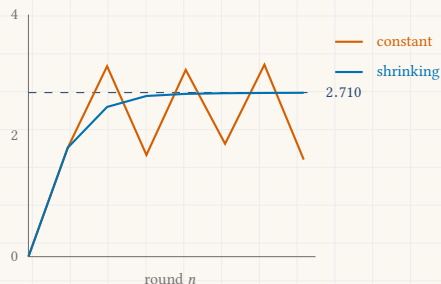


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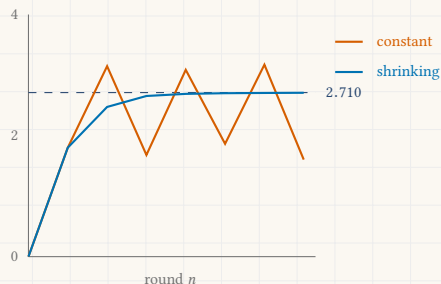


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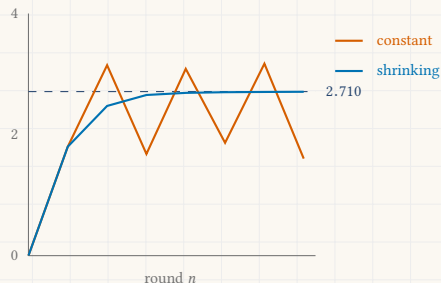
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The answer is $1 + \sqrt[3]{5} = 2.709976$.

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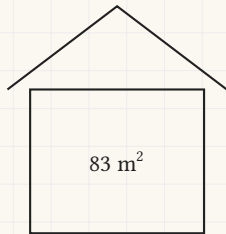
A step that never shrinks has to be lucky. A step that shrinks does not.

3

Section 3

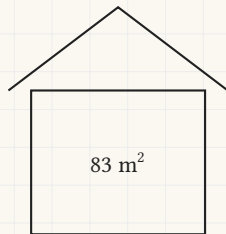
Estimating house prices

One house, one measurement



2 156 342 CHF

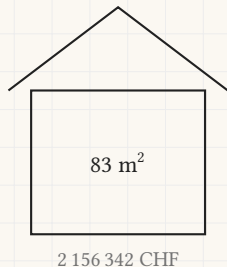
One house, one measurement



2 156 342 CHF

Data $x = 83$, the area. $y = 2\,156\,342$, the price.

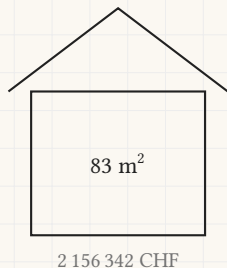
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Hypothesis Price is proportional to area, so the estimate is $\hat{y} = w x$.

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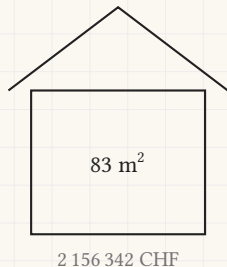


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Hypothesis Price is proportional to area, so the estimate is $\hat{y} = w x$.

Verdict The signed miss $l = y - \hat{y}$.

Problem Find the w that drives l to zero.

One row and one unknown, so the answer is $y/x = 25\,980.02$ francs per square metre, and the loop that follows never divides.

One house: the loop

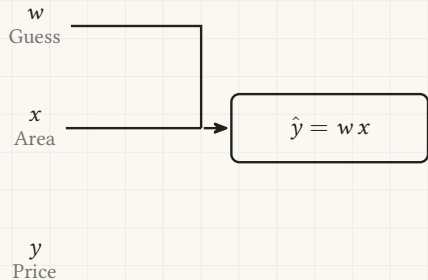
One house: the loop

w
Guess

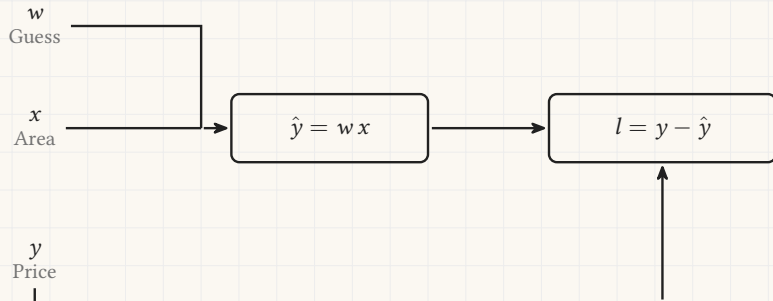
x
Area

y
Price

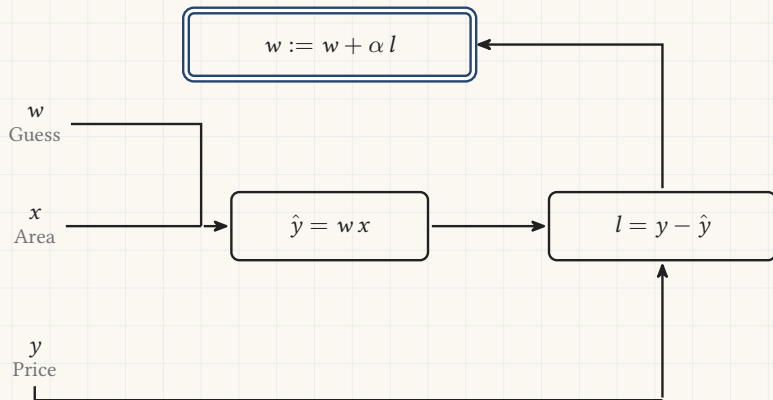
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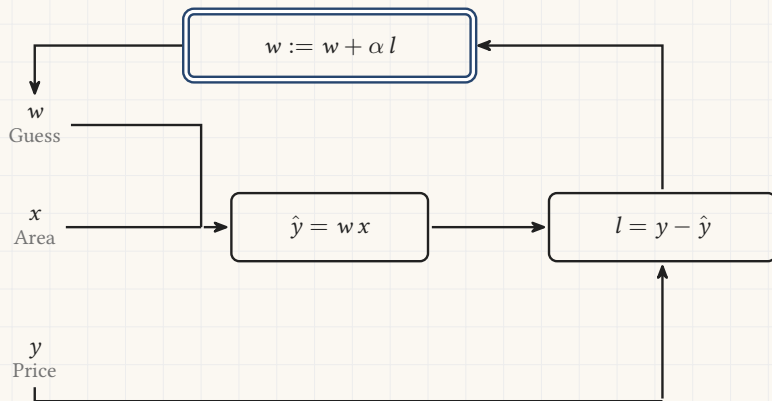
One house: the loop



One house: the loop



One house: the loop



One house: the algorithm

```
def fit(x, y, w, alpha):  
    l = y - w * x           # the verdict: y minus the estimate  
    while abs(l) > 0.5:      # more than half a franc out  
        w = w + alpha * l   # the update  
        l = y - w * x       # score the new guess  
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On the house above:

```
fit(83, 2156342, 20000, 0.01)
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```

On the house above:

```
fit(83, 2156342, 20000, 0.01)
```

Starting from a guess of 20 000 francs per square metre, this returns 25 980.02 after 8 rounds.

Many houses, one measurement

House	Area	Price
-------	------	-------

Many houses, one measurement

House	Area	Price
1	83	2 156 342
2	62	1 461 635
3	104	2 650 166
4	128	3 184 899
5	145	3 651 113

House 1 is the house on the last slide. Houses 2 to 5 are synthetic, generated with a fixed seed by `precompute/houses.py`.

Many houses, one measurement

Hypothesis The same one, row by row: $\hat{y}_i = w x_i$.

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Verdict One miss per row, $l_i = y_i - \hat{y}_i$.

Problem No single w empties every row. Drive the average verdict $\bar{l}$ to zero instead.

$$\bar{l} = \frac{l_1 + l_2 + \dots + l_n}{n}$$

House 1 is the house on the last slide. Houses 2 to 5 are synthetic, generated with a fixed seed by `precompute/houses.py`.

Many houses: the loop

Many houses: the loop

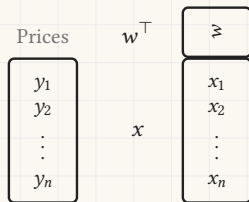
Prices

y_1
 y_2
 $\vdots$
 y_n

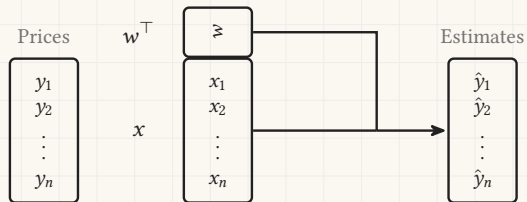
x

x_1
 x_2
 $\vdots$
 x_n

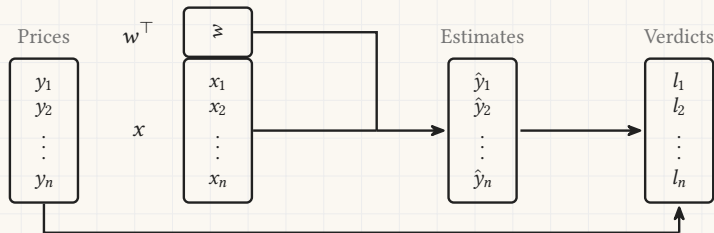
Many houses: the loop



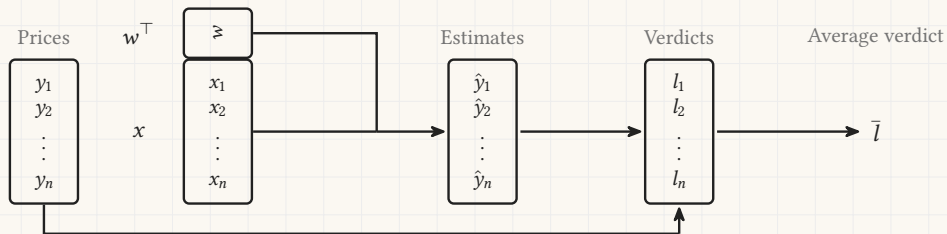
Many houses: the loop



Many houses: the loop

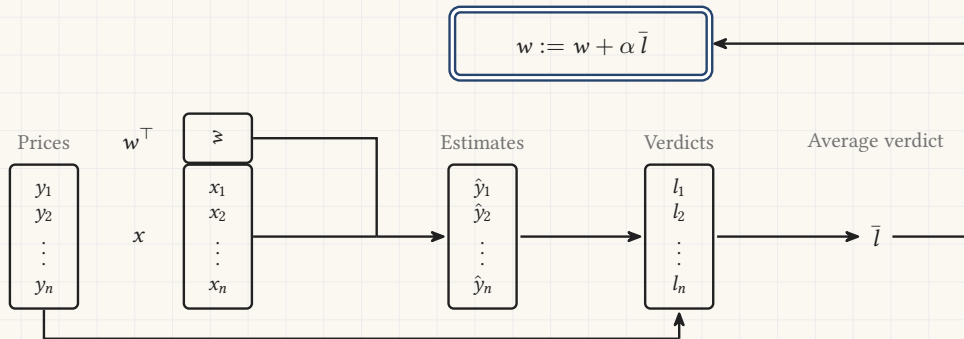


Many houses: the loop



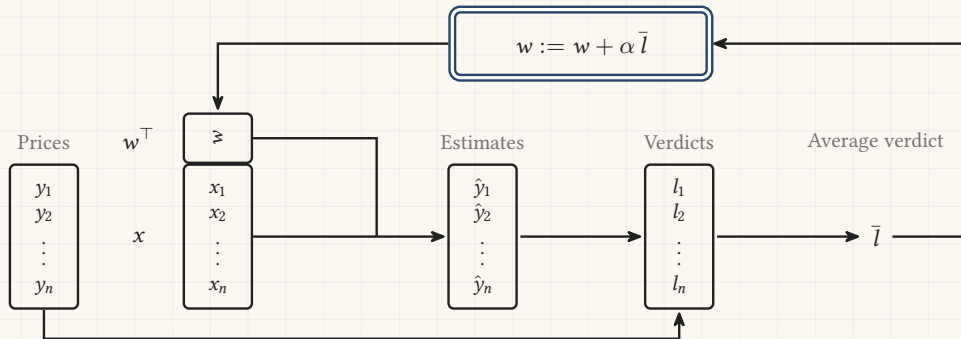
x holds the areas, and $\hat{y}_i = w x_i$, $l_i = y_i - \hat{y}_i$, $\bar{l} = (l_1 + \dots + l_n)/n$

Many houses: the loop



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Many houses: the loop



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Many houses: the algorithm

```
def verdict(x, y, w):           # the average miss over the rows
    total = 0
    for i in range(len(x)):
        total = total + (y[i] - w * x[i])
    return total / len(x)
def fit(x, y, w, alpha):
    l_bar = verdict(x, y, w)
    while abs(l_bar) > 0.5:
        w = w + alpha * l_bar
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The estimate, the verdict and the update are unchanged. Only the averaging is new.

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        l_bar = verdict(x, y, w)
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```

The estimate, the verdict and the update are unchanged. Only the averaging is new.

On the five houses, starting from 20 000 with $\alpha = 0.005$, this returns 25 103.74 after 19 rounds.

Two measurements for each house

House	Area	Rooms	Price
-------	------	-------	-------

Two measurements for each house

House	Area	Rooms	Price
1	83	3.5	2 156 342
2	62	2.5	1 461 635
3	104	4.5	2 650 166
4	128	5.5	3 184 899
5	145	6.5	3 651 113

Two measurements for each house

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1	83	3.5	2 156 342
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Hypothesis Each measurement gets its own weight:

$$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}.$$

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Verdict Still one number, $\bar{l}$, the average miss.

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Hypothesis Each measurement gets its own weight:
 $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}.$

Verdict Still one number, $\bar{l}$, the average miss.

Problem Two unknowns, one verdict. It does not say which weight is at fault.

So move one weight at a time.

Two measurements: the loop

Two measurements: the loop

Prices

y_1
y_2
$\vdots$
y_n

X

x_{11}	x_{12}
x_{21}	x_{22}
$\vdots$	$\vdots$
x_{n1}	x_{n2}

Two measurements: the loop

Prices

y_1
y_2
$\vdots$
y_n

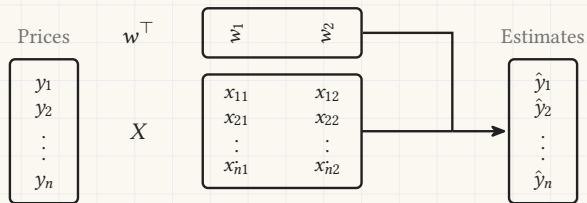
$w^\top$

w_1	w_2
-------	-------

X

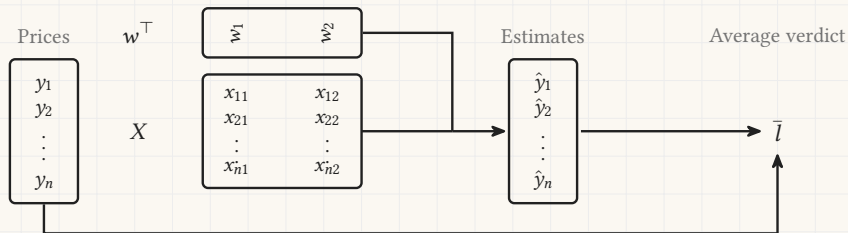
x_{11}	x_{12}
x_{21}	x_{22}
$\vdots$	$\vdots$
x_{n1}	x_{n2}

Two measurements: the loop



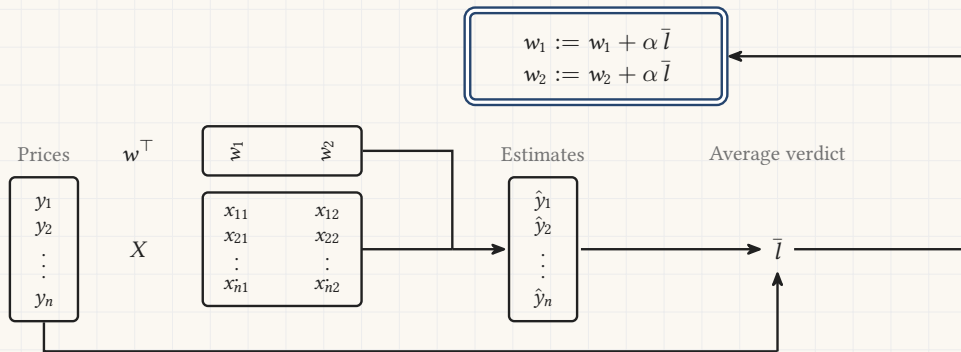
$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$, and the columns of X are area and rooms.

Two measurements: the loop



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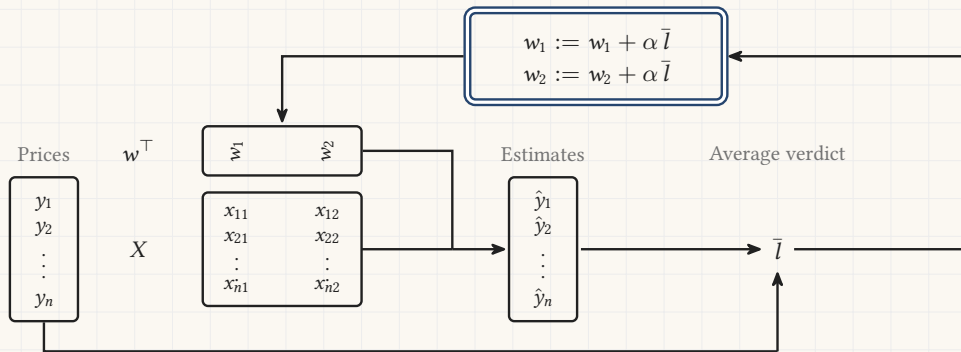
Two measurements: the loop



$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$, and the columns of X are area and rooms.

One weight moves per round, and they take turns.

Two measurements: the loop



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Two measurements: the algorithm

```
def verdict(x, y, w):
    total = 0
    for i in range(len(x)):
        total = total + (y[i] - w[0] * x[i][0] - w[1] * x[i][1])
    return total / len(x)

def fit(x, y, w, alpha):
    j = 0
    l_bar = verdict(x, y, w)
    while abs(l_bar) > 0.5:
        w[j] = w[j] + alpha * l_bar
        j = 1 - j                # take turns
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The loop alternates: area on the even rounds, rooms on the odd ones.

Two measurements: the algorithm

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```

The loop alternates: area on the even rounds, rooms on the odd ones.

Starting from $w = (20\,000, 0)$ with $\alpha = 0.005$, this returns $(25\,000.71, 2\,390.34)$ after 37 rounds.

Many measurements for each house

House	Area	Rooms	Age	Walk	Price

Many measurements for each house

House	Area	Rooms	Age	Walk	Price
1	83	3.5	12	4	2 156 342
2	62	2.5	41	11	1 461 635
3	104	4.5	28	8	2 650 166
4	128	5.5	6	15	3 184 899
5	145	6.5	55	6	3 651 113

Many measurements for each house

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- Each house is now a list of d measurements, $x_i = (x_{i1}, x_{i2}, \dots, x_{id})$.

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- Each house is now a list of d measurements, $x_i = (x_{i1}, x_{i2}, \dots, x_{id})$.
- Each measurement keeps its own weight, so $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$.

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House	Area	Rooms	Age	Walk	Price
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- Each house is now a list of d measurements, $x_i = (x_{i1}, x_{i2}, \dots, x_{id})$.
- Each measurement keeps its own weight, so $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$.
- One verdict, d unknowns. Pick one weight per round and move it.

Many measurements: the loop

Many measurements: the loop

Prices

y_1
y_2
$\vdots$
y_n

X

x_{11}	x_{12}	$\cdots$	x_{1d}
x_{21}	x_{22}	$\cdots$	x_{2d}
$\vdots$	$\vdots$	$\ddots$	$\vdots$
x_{n1}	x_{n2}	$\cdots$	x_{nd}

Many measurements: the loop

Prices

y_1
y_2
$\vdots$
y_n

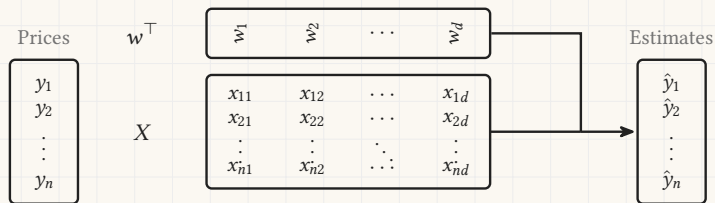
$w^\top$

w_1	w_2	$\dots$	w_d
-------	-------	---------	-------

X

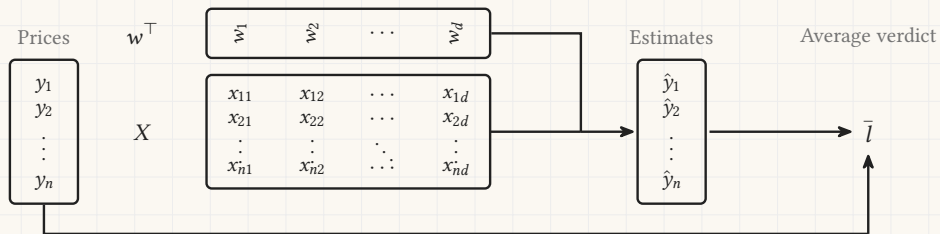
x_{11}	x_{12}	$\dots$	x_{1d}
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$\vdots$	$\vdots$	$\ddots$	$\vdots$
x_{n1}	x_{n2}	$\dots$	x_{nd}

Many measurements: the loop



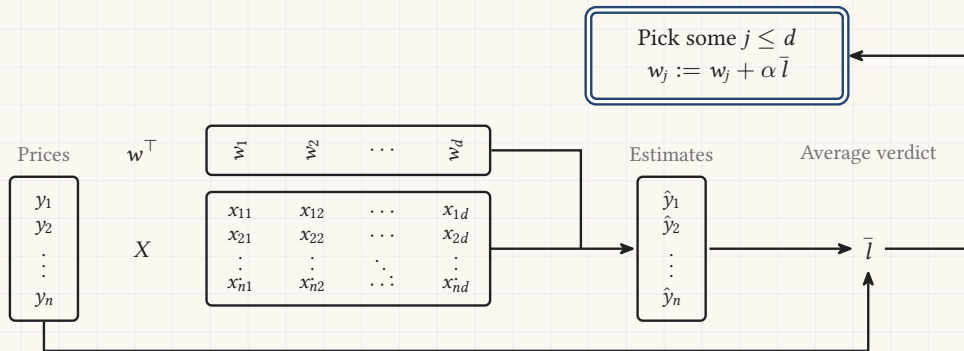
X has one row per house and one column per measurement, and $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$.

Many measurements: the loop



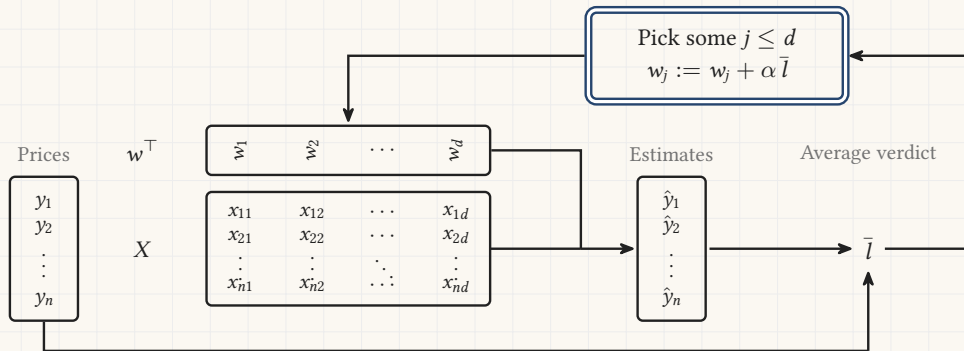
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Many measurements: the loop



X has one row per house and one column per measurement, and $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$.

The inner product

$$w^\top \begin{array}{|c|c|c|c|} \hline w_1 & w_2 & \dots & w_d \\ \hline \end{array}$$

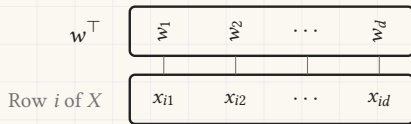
$$\begin{array}{|c|c|c|} \hline a & b & a^\top b \\ \hline \end{array}$$

The inner product

$$\begin{array}{c} w^\top \\ \text{Row } i \text{ of } X \end{array} \begin{array}{|c|c|c|c|} \hline w_1 & w_2 & \dots & w_d \\ \hline \end{array} \begin{array}{|c|c|c|c|} \hline x_{i1} & x_{i2} & \dots & x_{id} \\ \hline \end{array}$$

$$\begin{array}{|c|c|c|} \hline a & b & a^\top b \\ \hline \end{array}$$

The inner product

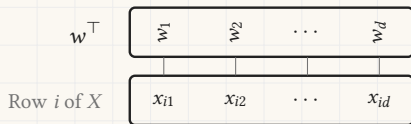


$$w^\top x_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$$

Multiply down each column, then add.

$$\begin{array}{r} \hline a \qquad b \quad a^\top b \\ \hline \\ \hline \end{array}$$

The inner product



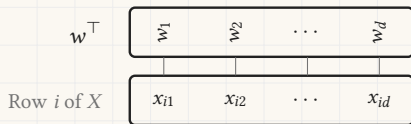
$$w^\top x_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$$

Multiply down each column, then add.

```
def inner(w, x):  
    total = 0  
    for j in range(len(w)):  
        total = total + w[j] * x[j]  
    return total
```

$$\begin{array}{ccc} a & b & a^\top b \end{array}$$

The inner product



$$w^\top x_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$$

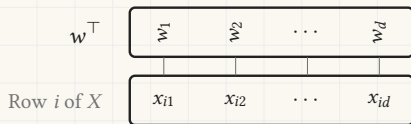
Multiply down each column, then add.

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def inner(w, x):  
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    return total
```

It also measures similarity.

a	b	$a^\top b$	
$(2, 1)$	$(2, 1)$	$+5$	The same direction

The inner product



$$w^\top x_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$$

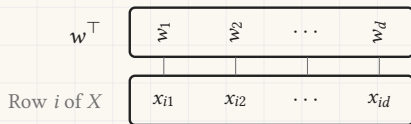
Multiply down each column, then add.

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    return total
```

It also measures similarity.

a	b	$a^\top b$	
$(2, 1)$	$(2, 1)$	$+5$	The same direction
$(2, 1)$	$(-1, 2)$	0	At right angles
$(2, 1)$	$(-2, -1)$	-5	Opposite directions

The inner product



$$w^T x_i = w_1 x_{i1} + w_2 x_{i2} + \dots + w_d x_{id}$$

Multiply down each column, then add.

```
def inner(w, x):  
    total = 0  
    for j in range(len(w)):  
        total = total + w[j] * x[j]  
    return total
```

It also measures similarity.

a	b	$a^T b$	
(2, 1)	(2, 1)	+5	The same direction
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It scores a document against a query, and one word against another in a language model. It returns all module.

Many measurements: the algorithm

```
def verdict(x, y, w):
    total = 0
    for i in range(len(x)):
        total = total + (y[i] - inner(w, x[i]))
    return total / len(x)

def fit(x, y, w, alpha):
    l_bar = verdict(x, y, w)
    while abs(l_bar) > 0.5:
        j = random.randrange(len(w))    # pick some j
        w[j] = w[j] + alpha * l_bar
        l_bar = verdict(x, y, w)
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        w[j] = w[j] + alpha * l_bar  
        l_bar = verdict(x, y, w)  
    return w
```

Starting from $w = (20\,000, 0, 0, 0)$ with $\alpha = 0.005$, this returns $(24\,566.95, 976.25, 1\,351.85, 1\,506.32)$ after 51 rounds.

Many measurements: the algorithm

```
def verdict(x, y, w):
    total = 0
    for i in range(len(x)):
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Starting from $w = (20\,000, 0, 0, 0)$ with $\alpha = 0.005$, this returns $(24\,566.95, 976.25, 1\,351.85, 1\,506.32)$ after 51 rounds.

The three algorithms before it are this one with $d = 1$ and $n = 1$, then $d = 1$, then $d = 2$.



Section 4

In AI

The training loop

The training loop

0

Data

The training loop

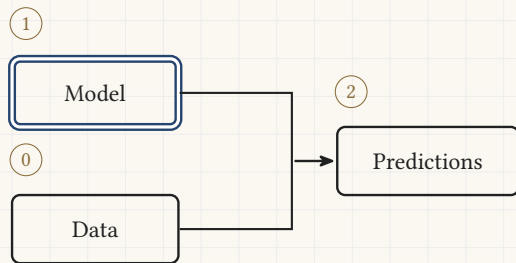
1

Model

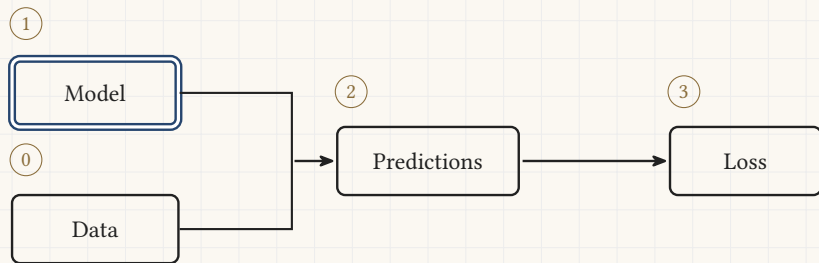
0

Data

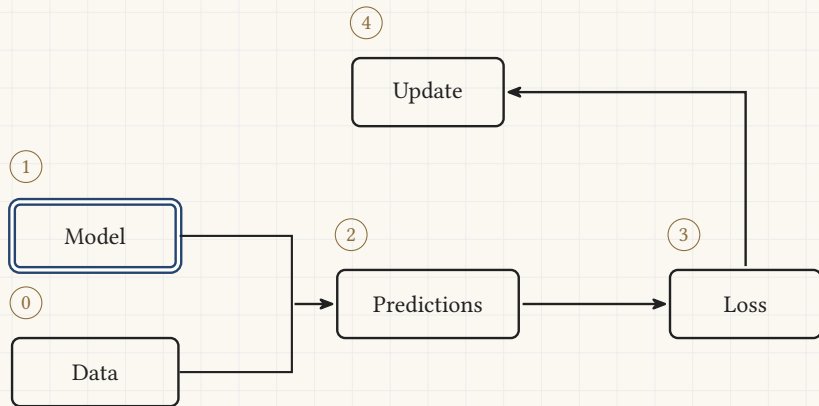
The training loop



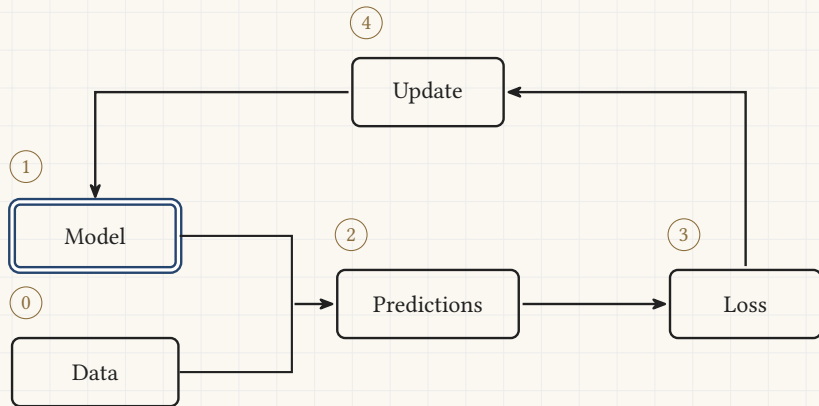
The training loop



The training loop



The training loop



Four boxes and a return arrow, run a few million times.

The morning's four words, and their AI names

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Guess $\rightarrow$ Model
The weights w

The morning's four words, and their AI names

Guess → Model

The weights w

Estimate → Predictions

What the weights say about the data

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Verdict → Loss

How wrong, and in which direction

The morning's four words, and their AI names

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The weights w

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What the weights say about the data

Verdict → Loss

How wrong, and in which direction

Update → Training step

$w := w + \alpha l$

Statistical learning theory

Plate not present

`img/vapnik.jpg`

See `img/IMAGE-SOURCES.txt`

Vladimir Vapnik



Statistical learning theory

Plate not present
img/vapnik.jpg
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- All morning we moved w until the miss on *our* rows was small.



Statistical learning theory

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Statistical learning theory

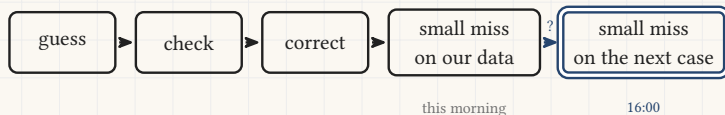
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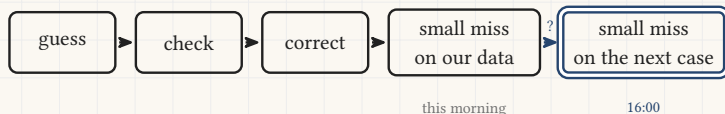
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Vapnik prices that arrow: the miss on the next case is at most the miss on our data, plus a penalty that grows with how much the family can bend and shrinks as rows are added. A price rather than a promise.

Weekend 1, Friday morning

5

Section 5

Where this ends up

Modern applications

Most modern machine learning and AI applications are first-order approximation algorithms.

Family

What its verdict compares

Modern applications

Most modern machine learning and AI applications are first-order approximation algorithms.

Family	What its verdict compares
Large language models	The next token against the token that followed

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Four different verdicts. One update rule, and it is the one on the square-root table.